WORKSHEET

Rate problems

1 Simone assembles conference folders. She receives $1.20 for each folder she assembles. How much will Simone receive for assembling 360 folders?

2 Karen has a cleaning business. She charges $10 for travel plus $20 per hour.

a How much will Karen charge to travel to a house and clean for 5 hours?

b Karen charges $70 to clean an office. How long did the cleaning take?

3 Ben cleans pools. He charges $60 for a service call, which includes the first 30 minutes. He charges $10 for every 15 minutes after that. How much does Ben charge for a 1 hour service call?

4 Alison has a part-time job ironing clothes. If the ironing is dropped off and picked up by the customer, she charges $3 per item. If she has to pick up and deliver, she charges an additional $20.

a How much does Alison charge a customer who delivers and picks up 40 items?

b Alison has to pick up and return a basket of 36 items. How much does she charge the customer?

5 Jessica likes to knit. If she is doing plain knit stitch, she can complete 24 stitches per minute.

a How many stitches can Jessica complete during a 30-minute TV show?

b Jessica is making a scarf. The scarf is 36 stitches wide and 160 rows long. How long will it take Jessica to knit the scarf?

6 Graham is cutting up fence palings to use as firewood. He cuts up 120 palings in 40 minutes. In palings per minute, calculate the rate Graham is cutting up palings.

© Cengage Learning Australia Pty Ltd 2014

7 In one particular fortnight over summer, surf lifesavers made 294 rescues. Calculate this rate in rescues per day.

8 Bill and Peter both own nail guns. Bill's gun can fire nails at a rate of 25 nails per 30 seconds while Peter's gun fires at a rate of 50 nails per minute. Whose gun can fire at the faster rate? Justify your answer.

9 Carol is taking 2 painkillers ever 6 hours to relieve a toothache.

a How many tablets is Carolyn taking every day (24 hours)?

b At this rate, how long will a box containing 24 tablets last?

10 Over a 3-day long weekend, police checked the speed of 12 960 motorists travelling on a highway. They charged 5% of the total motorists checked with speeding offences. In drivers charged per hour, calculate the rate at which the police were charging drivers with speeding offences.

© Cengage Learning Australia Pty Ltd 2014

Answers

1 $432

2 a $110

b 3.5 hours

3 $80

4 a $120

b $128

5 a 720

b 240 minutes = 4 hours

6 3 palings / minute

7 21 rescues / day

8 They both fire at the same rate of 50 nails / minute.

9 a 8

b 3 days

10 9 drivers / hour

© Cengage Learning Australia Pty Ltd 2014